



Effects of dietary supplementation of enzymatic bio-conversion of *Scutellaria baicalensis* extract as an alternative to antibiotics on the growth performance, nutrient digestibility, fecal microbiota, fecal gas emission, blood hematology, and antioxidative indicators in growing pigs

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HIGHLIGHTS

- Enzymatic bio-conversion of SB extract supplementation showed tendency to improve growth performance.
- The GSH level was increased in pigs treated with enzymatic bio-conversion of SB extract.
- 0.1% SB extract added in diet did not affect the performance of pigs.
- Positive correlative between enzymatic bio-conversion of SB extract supplementation in the diet and ADG and GSH and GPX concentration was observed.

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ABSTRACT

This study was carried out to assess the effects of dietary supplementation of enzymatic bio-conversion of *Scutellaria baicalensis* (SB) extract on the growth performance, nutrient digestibility, fecal microbiota, fecal gas emission, blood hematology, and antioxidative indicators in growing pigs. A total of 200 crossbred pigs [(Landrace × Yorkshire) × Duroc] with an initial body weight of 24.52 ± 1.99 kg were used in a 42-day experiment with a completely randomized block design. Pigs were randomly divided into 4 treatment groups with 10 replicate pens per treatment and 5 animals per pen (2 barrows and 3 gilts). There were four diet conditions: 1) CON, basal diet; 2) TRT1, basal diet + 0.1% apramycin; 3) TRT2, basal diet + 0.1% SB extract; 4) TRT3, basal diet + 0.1% enzymatic bio-conversion of SB extract. Compared with the CON group, TRT1 or TRT3 groups had higher final body weight (BW) ($P = 0.074$) and average daily gain (ADG) ($P = 0.075$), and lower feed conversion ratio (FCR) ($P = 0.054$). The apparent total tract digestibility (ATTD) of dry matter (DM) ($P = 0.095$) in TRT1 group was higher than that in CON group. Serum glutathione peroxidase (GPX) ($P = 0.047$) and glutathione (GSH) ($P = 0.074$) concentrations in TRT3 group were higher than those in CON or TRT1 groups. However, there were no significant differences in fecal microbiota, fecal gas emission, and blood hematology among all dietary groups. A positive correlation between apramycin supplementation in the diet and ADG and the ATTD of DM were observed. A positive correlation between enzymatic bio-conversion of SB extract supplementation in the diet and ADG and the concentration of GPX and GSH in serum were also observed. As a result, the enzymatic bio-conversion of SB extract has the potential to be used as an alternative to antibiotics due to the comparable effects with apramycin on the growth performance, nutrient digestibility, blood hematology, fecal microbiota, and fecal gas emission, moreover, the enzymatic bio-conversion of SB extract was better than apramycin in the improvement of antioxidative indicators.

1. Introduction

Antibiotic growth promoters (AGP) have been used in swine

production for more than five decades to improve the performance and health of animals (Allen et al., 2013). In recent years, it was recognized that the use of AGP in animal feed resulted in antimicrobial resistance

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issues eventually threatening safe animal production as well as human health (Byarugaba, 2004). Therefore, seeking viable antibiotic alternatives is getting considerable attention for the improvement of the livestock industry. Particular interest is now being paid to the phytogetic feed additives due to their growth-promoting effects and health benefits (Ao et al., 2011; Lan et al., 2016). Phytogetic feed additives have been reported to be safe and environmentally friendly and have been considered as alternatives to antibiotics (Li et al., 2015b) due to their positively affected growth performance, nutrient digestibility, intestinal microbiota, meat quality, fecal gas emission, and oxidative situation in pigs (Lei et al., 2018; Chen et al., 2019).

Scutellaria baicalensis (Skullcap; Huangqi; Labiatae family; SB) is a traditional herb with high pharmacological and therapeutic effects attributed to flavonoids that has been widely studied (Rodriguez et al., 2017). Traditionally, it has been used to treat bacterial and viral infections of the respiratory tract and gastrointestinal tract, cardiovascular disease and inflammation (Thanh et al., 2015; Zhao et al., 2016b). Moreover, the SB also has antioxidant and free-radical scavenging properties (Cao et al., 2011). When used in swine husbandry, Huang et al. (2019) noted that dietary supplementation of 1000 mg/kg SB extract positively affects intestinal morphology and barrier function through restraining NF- κ B/P38 pathways to reduce the diarrhea rate in weaning pigs. Liu et al. (2016); Zhao et al. (2016a); Lei et al. (2018) reported that supplementation of herbal complexes extract which containing SB extract in the diet of pigs positively affects growth performance, nutrient digestibility, cholesterol content in blood, fecal noxious gas emission, meat quality, and diarrhea condition. The factors of positive effects of dietary supplement SB extract were considered as flavonoids in the SB (Zhou et al., 2019).

Recently, the method of enzymatic bio-conversion of SB extract to enhance the bio-availability and functional value has been reported (Xu et al., 2013; Rodriguez et al., 2017). Biotransformation refers to the use of a pure enzyme or a whole cell culture for the structural transformation of molecules (Zafar et al., 2016). Enzymes being specific in nature can catalyze the reaction at specific sites in the molecules (Zafar et al., 2016). The enzymatic bio-conversion of SB extract could increase the effective compound (Rodriguez et al., 2017) and consequently improve its therapeutic efficacy (Macedo et al., 2011; Bhullar et al., 2013).

However, no studies have been conducted to supplement enzymatic bio-conversion of SB extract in the diet of animals, moreover, studies on the pigs consuming an SB extract containing diet are also limited. As the above description, the SB extract or enzymatic bio-conversion of SB extract probably benefited the performance of animals. Thus, the objective of this study was to test the hypothesis that a diet supplemented with SB extract or enzymatic bio-conversion of SB extract positive affect the growth performance, nutrient digestibility, fecal microbiota, fecal gas emission, blood hematology, and antioxidative indicators in growing pigs, moreover, the enzymatic bio-conversion of SB extract has more potential to be used as an alternative to antibiotics than the SB extract.

2. Material and methods

The protocol of this experiment was approved by the Animal Care and Use Committee of Dankook University (Approval, No: DK-2-1808).

2.1. Preparation of the additive

The commercial enzymatic bio-conversion of SB extract used in this study was obtained from Biogenoci Co., Ltd. (Suwon, Korea). Based on the manufacturer information, the *Aspergillus niger* A-T1 (Korean Culture Center of Microorganisms 12147P) was first cultivated into PDA medium (potato extract 4 g, dextrose 20 g, agar 15 g/liter) for 5 days at 30 °C. After culture, *Aspergillus niger* A-T1 was inoculated into sterilized enzyme induction media (5.0×10^6 cfu/g wet weight), mixed with 30% wheat bran, 20% *Scutellaria baicalensis* powder, and 50% tap

water, cultured at 30 °C for 4 days, and dried to prepare induction enzyme (β -glucuronidase). Then, 15% *Scutellaria baicalensis* powder was soaked in tap water that treated with acetic acid to pH 4.5, heated for 30 min at 105 °C, and then cooled to 40 °C. 5% dried induction enzyme supplement into 15% *Scutellaria baicalensis* dissolved solution (pH 4.5) and reaction (bio-conversion) 6 h at 40 °C and 25 rpm. After reaction, drying reacted solution by heat dryer and milling to take the enzymatic bio-conversion of *Scutellaria baicalensis* extract.

The induction enzyme (β -glucuronidase) could bio-transform of baicalin into baicalein and wogonoside into wogonin (Kim et al., 2009; Xu and Ji, 2013), thereby improving the biological activity of flavonoids in SB (Guo et al., 2011).

2.2. Animals and housing

A total of 200 growing pigs [(Landrace $\times$ Yorkshire) $\times$ Duroc] with an initial body weight of 24.52 ± 1.99 kg were used in a randomized complete block design experiment. All pigs were housed in an environmentally controlled room, which was thermostatically controlled and maintained at 25 °C, throughout the experiment. The floor of the barn was slatted plastic floor. Pens were equipped with a one-sided feeder and nipple drinker.

2.3. Experimental design and diets

Based on the initial body weight, pigs were assigned to 4 treatments, each treatment had 10 replicate pens and 5 pigs (2 barrows and 3 gilts) per pen. In this 42-d experiment, pigs were fed with basal diet (CON), and a basal diet supplemented with 0.1% apramycin (TRT1), 0.1% SB extract (TRT2), or 0.1% enzymatic bio-conversion of SB extract (TRT3). Each of the treatments was added to the basal diet at the expense of corn and all diets were formulated to meet the nutrient requirements recommended by the National Research Council (NRC 2012; Table 1). All pigs had free access to feed and water throughout the experimental period.

2.4. Sample collection and measurements

Individual body weight and feed intake on pen basis were recorded at the beginning and the end of the experiment to calculate the average daily gain (ADG), average daily feed intake (ADFI), and feed conversion ratio (FCR). On the 35th day of the experiment, 0.2% chromium oxide (Cr_2O_3) was added to these diets as an indigestible marker to determine the apparent total tract digestibility (ATTD) of dry matter (DM), nitrogen (N), and energy. After that, the representative feed samples were collected from each dietary treatment. On day 41, two pigs were randomly selected from each pen to take the fecal samples via the rectal massage method. All feed and fecal samples were stored at -20 °C until analysis. Before chemical analysis, samples were oven-dried at 60 °C for 72 h and then ground into powder, which can through a 1-mm sieve. All feed and fecal samples were analyzed using the methods of AOAC (2007) to measure the DM (method 930.15) and N (method 984.13). The combustion heat was measured by a bomb calorimeter (Parr 6100; Parr Instrument Co., Moline, IL, USA) to determine the energy. The chromium levels were analyzed via UV absorption spectrophotometry (UV-1201, Shimadzu, Kyoto, Japan). The ATTD was then calculated using the following formula: $\text{ATTD}(\%) = [1 - \{(\text{Nf} \times \text{Cd}) / (\text{Nd} \times \text{Cf})\}] \times 100$, where Nf = nutrient concentration in feces (% DM), Nd = nutrient concentration in diet (% DM), Cd = chromium concentration in diet (% DM), and Cf = chromium concentration in feces (% DM).

On day 39, the method of rectal massage was used to take fecal samples from two randomly selected pigs in each pen for measuring the fecal ammonia (NH_3), hydrogen sulfide (H_2S), and total mercaptans (R-SH) emission. The fecal samples from the same pen were pooled, mixed, and transported to lab for further analysis of fecal noxious gas emission. For analysis of noxious gas emission, the fecal samples were stored in

Table 1
Composition of growing pig diets (as fed-basis).

Item	Dietary treatment ¹			
	CON	TRT1	TRT2	TRT3
Ingredients				
Corn	73.99	73.78	73.78	73.78
Soybean meal	21.31	21.34	21.34	21.34
Tallow (liquid)	1.78	1.86	1.86	1.86
Dicalcium phosphate	1.24	1.24	1.24	1.24
Limestone	0.75	0.75	0.75	0.75
Salt	0.20	0.20	0.20	0.20
Methionine	0.06	0.06	0.06	0.06
Lysine	0.42	0.42	0.42	0.42
Mineral premix ²	0.10	0.10	0.10	0.10
Vitamin premix ³	0.12	0.12	0.12	0.12
Choline	0.03	0.03	0.03	0.03
Apramycin	–	0.10	–	–
Scutellaria baicalensis (SB) extract	–	–	0.10	–
Enzymatic bio-conversion of SB extract	–	–	–	0.10
Total	100.00	100.00	100.00	100.00
Calculated value				
Metabolism energy, kcal/kg	3300	3300	3300	3300
Crude protein,%	16.50	16.50	16.50	16.50
Crude fat,%	4.64	4.71	4.71	4.71
Calcium,%	0.66	0.66	0.66	0.66
Total phosphorus,%	0.56	0.56	0.56	0.56
Total lysine,%	1.12	1.12	1.12	1.12
Total methionine,%	0.32	0.32	0.32	0.32

¹ Dietary treatments were as follows: CON, basal diet; TRT1, basal diet + 0.1% apramycin; TRT2, basal diet + 0.1% SB extract; TRT3, basal diet + 0.1% enzymatic bio-conversion of SB extract.

² Provided per kg diet: Fe, 100 mg as ferrous sulfate; Cu, 17 mg as copper sulfate; Mn, 17 mg as manganese oxide; Zn, 100 mg as zinc oxide; I, 0.5 mg as potassium iodide; and Se, 0.3 mg as sodium selenite.

³ Provided per kilograms of diet: vitamin A, 10,800 IU; vitamin D₃, 4000 IU; vitamin E, 40 IU; vitamin K₃, 4 mg; vitamin B₁, 6 mg; vitamin B₂, 12 mg; vitamin B₆, 6 mg; vitamin B₁₂, 0.05 mg; biotin, 0.2 mg; folic acid, 2 mg; niacin, 50 mg; D-calcium pantothenate, 25 mg.

2.6-L sealed plastic boxes in duplicates. Each box had a small hole in the middle of one sidewall that was sealed by adhesive plaster. The samples were fermented in the sealed boxes for 24 h at 25 °C. Before doing the measurement, the fecal samples were shaken manually for 30 s for homogenizing. A gas-sampling pump (model GV-100S; Gastec Corp., Japan; Gastec detector tube No.3La for ammonia; No. 4LK for hydrogen sulfide; No. 70 L for total mercaptans) was used to analyze the gas sample. The head-space air was sampled approximately 2.0 cm above the surface of excreta through the small hole. Two samples from each pen were measured and then the average was calculated.

On the last day of the experiment, two pigs per pen were selected randomly and were bled for collecting blood samples via anterior vena cava puncture into non-heparinized vacuum tubes (Becton Dickinson Vacutainer Systems, Franklin Lakes, NJ, USA). The blood samples were collected during 11:00 to 12:00 h in order to exclude the circadian fluctuations in hormone concentrations. Blood samples were centrifuged (3000 × g) for 15 min at 4 °C to obtain serum samples and then stored at –20 °C until analysis. Serum cortisol concentration was analyzed by solid-phase cortisol RIA kit (Coat-A-Count TKCO, Diagnostic Products Corporation, Los Angeles, CA, USA). The epinephrine and norepinephrine levels were quantified using an ion-exchange purification procedure followed by high-performance liquid chromatography with electrochemical detection. The intra- and inter-assay coefficients of variation were 7.0% and 7.1% for norepinephrine and 6.5% and 11.6% for epinephrine, respectively. The concentration of blood urea nitrogen (BUN), creatinine, protein, and insulin in serum were analyzed by an automatic biochemical blood analyzer (HITACHI 747; Hitachi, Tokyo, Japan). The serum antioxidant activity of glutathione peroxidase (GPX), glutathione (GSH), and glutathione disulfide (GSSG) were detected using the ELISA kit (R & D Systems, Minneapolis, MN, USA).

At the end of the experiment, fecal samples were collected for

measuring fecal *E. coli* and *Lactobacillus* counts via massaging the rectum from two random pigs from each pen and then pooled and placed on ice for transportation to the laboratory, where analysis was immediately carried out. One gram of fecal sample was diluted with 9 ml of 1% peptone broth (Becton, Dickinson, Franklin Lakes, NJ, USA) and then homogenized. Viable bacteria in the fecal samples were then counted by plating serial 10-fold dilutions (in 1% peptone solution) onto MacConkey agar plates (Difco Laboratories, Detroit, MI, USA) and the Lactobacilli medium III agar plates (Medium 638, DSMZ, Braunschweig, Germany) in duplicate to isolate the *E. coli* and *Lactobacillus*, respectively. The lactobacilli medium III agar plates were then incubated for 48 h at 39 °C under anaerobic conditions. The MacConkey agar plates were incubated for 24 h at 37 °C under aerobic conditions. The *E. coli* and *Lactobacillus* colonies were counted immediately after removal from the incubator.

2.5. Statistical analysis

Data were analyzed using the GLM procedure of SAS (SAS Inst. Inc., Cary, NC, USA) as randomized complete block design. The pen was taken as the experiment unit. Tukey's multiple range test was used for evaluation of differences among the treatments. Variability in the data was expressed as the standard error of means (SBM) and $P < 0.05$ was considered to be statistically and $P < 0.10$ was considered as a trend. Pearson's correlation coefficients and probabilities were generated using the Correlation procedure of SAS.

3. Results

The increased tendency regarding final body weight ($P = 0.074$) and average daily gain ($P = 0.075$), and the decreased tendency regarding feed conversion ratio ($P = 0.054$), were observed in pigs consuming TRT1, TRT2, or TRT3 group diets. Moreover, the final body weight and average daily gain in the TRT1 or TRT3 groups were higher than CON group, besides, the feed conversion ratio in the TRT1 or TRT3 groups was lower than CON group. However, there was no statistical difference in average daily feed intake between all dietary groups (Table 2).

The increased tendency regarding apparent total tract digestibility of dry matter ($P = 0.095$) was observed in pigs consuming TRT1, TRT2, or

Table 2
Effects of dietary supplementation of enzymatic bio-conversion of *Scutellaria baicalensis* (SB) extract on growth performance and nutrient digestibility in growing pigs^{1,2}.

Items	CON	TRT1	TRT2	TRT3	SEM	P-value
Body weight, kg						
Initial	24.517	24.518	24.524	24.523	0.003	0.254
Final	53.082 ^b	54.557 ^a	53.761 ^{ab}	54.458 ^a	0.428	0.074
Growth performance						
ADG, g	680.120 ^b	715.220 ^a	696.110 ^{ab}	712.740 ^a	10.180	0.075
ADFI, g	1489.100	1522.430	1501.860	1522.190	16.642	0.424
FCR	2.190 ^a	2.129 ^b	2.159 ^{ab}	2.138 ^b	0.016	0.054
Apparent total tract digestibility,%						
DM	76.761 ^b	78.409 ^a	77.954 ^{ab}	77.691 ^{ab}	0.447	0.095
N	75.288	76.790	76.211	75.860	0.608	0.381
Energy	76.475	77.414	77.135	76.970	0.362	0.340

Abbreviation: ADG, average daily gain; ADFI, average daily feed intake; FCR, feed conversion ratio; DM, dry matter; N, nitrogen; SEM, standard error of the mean.

¹Dietary treatments were: CON, basal diet; TRT1, basal diet + 0.1% apramycin; TRT2, basal diet + 0.1% SB extract; TRT3, basal diet + 0.1% enzymatic bio-conversion of SB extract.

²Values represent the means of five pens with five pigs per replicate pens ($n = 25$) per treatment for body weight and five pens ($n = 5$) per treatment for ADG, ADFI, FCR. Moreover, two pigs per replicate pens ($n = 10$) per treatment for the apparent total tract digestibility of DM, N, and energy.

^{a,b}Different superscripts within a row indicate a significant difference ($P < 0.05$).

TRT3 group diets. Besides, the apparent total tract digestibility of dry matter in the TRT1 group was higher than CON group. The apparent total tract digestibility of nitrogen and the energy retention were not affected by pigs consuming TRT1, TRT2, or TRT3 group diets (Table 2).

The concentration of protein, insulin, blood urea nitrogen, creatinine, cortisol, epinephrine, norepinephrine, and glutathione disulfide were insignificantly affected by pigs consuming TRT1, TRT2, or TRT3 group diets (Table 3). However, the concentration of serum glutathione peroxidase ($P = 0.047$) was increased in pigs consuming TRT1, TRT2, or TRT3 group diets, besides, an increased tendency about the concentration of serum glutathione ($P = 0.074$) was also observed. Moreover, the concentration of serum glutathione peroxidase and glutathione in pigs consuming TRT3 group diet were higher than those consuming CON or TRT1 group diets.

The fecal microbiota (*Lactobacillus* and *E. coli*) and fecal gas emission (ammonia, total mercaptan, and hydrogen sulfide) were not affected by pigs consuming TRT1, TRT2, or TRT3 group diets (Table 4).

The Pearson correlation coefficients and probabilities between the apparent total tract digestibility of dry matter, final body weight, average daily gain, feed conversion ratio and the supplementation of apramycin in the diet are presented in Table 5. A strong negative correlation was seen between supplementation of apramycin in the diet and feed conversion ratio. Final body weight had a strong positive correlation with average daily gain. Average daily gain had a strong negative correlation with feed conversion ratio. The supplementation of apramycin in the diet positively correlated with average daily gain and the apparent total tract digestibility of dry matter.

The Pearson correlation coefficients and probabilities among serum glutathione peroxidase and glutathione concentrations, final body weight, average daily gain, feed conversion ratio, and the supplementation of enzymatic bio-conversion of SB extract in the diet are presented in Table 6. A strong negative correlation was seen between supplementation of enzymatic bio-conversion of SB extract in the diet and feed conversion ratio. Final body weight had a strong positive correlation with average daily gain. Average daily gain had a strong negative correlation with feed conversion ratio. Final body weight had a negative correlation with feed conversion ratio. The supplementation of enzymatic bio-conversion of SB extract in the diet positively correlated with average daily gain and the concentrations of glutathione in serum. In addition, supplementation of enzymatic bio-conversion of SB extract in the diet had a strong positive correlation with the concentration of glutathione peroxidase in serum.

Table 3

Effects of dietary supplementation of enzymatic bio-conversion of *Scutellaria baicalensis* (SB) extract on blood hematology and antioxidative indicators in growing pigs^{1,2}.

Items	CON	TRT1	TRT2	TRT3	SEM	P-value
Blood hematology						
Protein, g/dL	6.625	6.500	6.400	6.475	0.160	0.796
Insulin, U/mL	26.000	33.750	32.000	28.000	2.458	0.171
BUN, mg/dL	12.500	13.750	13.250	13.000	1.397	0.934
Creatinine, mg/dL	0.888	1.040	0.963	0.898	0.045	0.129
Cortisol, ug/dL	1.680	1.380	1.085	1.188	0.180	0.170
Epinephrine, ug/dL	1.230	1.130	0.983	0.945	0.085	0.136
Norepinephrine, ug/dL	2.113	2.050	1.873	1.933	0.093	0.313
Antioxidative indicators						
GPX, nmol/min/mL	77.031 ^b	78.713 ^b	82.801 ^{ab}	85.229 ^a	1.884	0.047
GSH, μ mol/L	0.007 ^b	0.007 ^b	0.009 ^{ab}	0.011 ^a	0.001	0.074
GSSG, nmol/mg	1.998	1.940	1.858	1.853	0.114	0.775

Abbreviation: BUN, blood urea nitrogen; GPX, glutathione peroxidase; GSH, glutathione; GSSG, glutathione disulfide; SEM, standard error of the mean.

¹Dietary treatments were: CON, basal diet; TRT1, basal diet + 0.1% apramycin; TRT2, basal diet + 0.1% SB extract; TRT3, basal diet + 0.1% enzymatic bio-conversion of SB extract.

²Values represent the means of five pens with two pigs per replicate pen ($n = 10$) per treatment.

^{a,b}Different superscripts within a row indicate a significant difference ($P < 0.05$).

Table 4

Effect of dietary supplementation of enzymatic bio-conversion of *Scutellaria baicalensis* (SB) extract on fecal microbiota and fecal gas emission in growing pigs¹.

Items	CON	TRT1	TRT2	TRT3	SEM	P-value
Fecal microbiota counts, log ₁₀ cfu g ⁻¹						
<i>Lactobacillus</i>	7.267	7.360	7.319	7.307	0.061	0.756
<i>E. coli</i>	6.306	6.244	6.271	6.265	0.055	0.877
Noxious gas emission, ppm						
H ₂ S	4.300	2.975	3.350	3.825	0.398	0.172
NH ₃	12.350	10.800	11.350	11.075	1.102	0.772
R-SH	1.100	0.700	0.750	0.900	0.204	0.537

Abbreviation: H₂S, hydrogen sulfide; NH₃, ammonia; R-SH, total mercaptan; SEM, standard error of the mean.

¹Dietary treatments were: CON, basal diet; TRT1, basal diet + 0.1% apramycin; TRT2, basal diet + 0.1% SB extract; TRT3, basal diet + 0.1% enzymatic bio-conversion of SB extract.

²Values represent the means of five pens with 2 pigs per replicate pen ($n = 10$) per treatment.

Table 5

Pearson correlation coefficients between apramycin supplementation in the diet and final body weight (BW), average daily gain (ADG), feed conversion ratio (FCR) and between apramycin and apparent total tract digestibility of dry matter (DM).

Variables	Apramycin	Final BW	ADG	FCR
Final BW	0.328	1.000		
ADG	0.540*	0.873***	1.000	
FCR	-0.658**	-0.406	-0.612**	1.000
Variables	Apramycin	DM		
DM	0.507*	1.000		

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

4. Discussion

In the present study, supplementation of SB extract in the diet of pigs did not affect growth performance. Similar results were reported by Huang et al. (2019), they noted that weaned piglets challenged with *E. coli* K 88 and consuming a 0.1% SB extract containing diet did not affect the final body weight and average daily gain. Liu et al. (2016) reported that finishing pigs fed a diet supplemented with 0.025 or 0.05% herbal extract mixture, which contained SB extract and other herbal extract did not affect the average daily feed intake. Studies on the supplementation

Table 6

Pearson correlation coefficients between enzymatic bio-conversion of *Scutellaria baicalensis* (SB) extract supplementation in the diet and final body weight (BW), average daily gain (ADG), feed conversion ratio (FCR) and between enzymatic bio-conversion of SB extract and the concentration of glutathione peroxidase (GPX) and glutathione (GSH).

Variables	Enzymatic bio-conversion of SB extract	Final BW	ADG	FCR
Final BW	0.330	1.000		
ADG	0.456*	0.806***	1.000	
FCR	-0.569**	-0.464*	-0.724***	1.000
Variables	Enzymatic bio-conversion of SB extract	GPX	GSH	
GPX	0.658**	1.000		
GSH	0.610*	0.242	1.000	

* $P < 0.05$.

** $P < 0.01$.

*** $P < 0.001$.

of SB extract in the diet of pigs were limited, without having other studies for comparison with this study. Dietary supplementation of SB extract insignificantly affects the growth performance of pigs probably due to the low bioavailability of flavonoids from SB extract when animals consuming SB extract containing diet (Zhang et al., 2013). The baicalein (Vergun et al., 2019), baicalin (Kim et al., 2019), wogonin (Liu et al., 2016), and wogonoside (Zhao et al., 2016b) were major flavonoids compounds from the SB extract, besides, the bioavailability of baicalein or wogonin were higher than baicalin or wogonoside, respectively (Lai et al., 2003; Zhang et al., 2005). In this study, the induction enzyme of β -glucuronidase was used to react with SB extract, it can bio-transform of baicalin into baicalein and wogonoside into wogonin (Kim et al., 2009; Xu and Ji, 2013). Study of Zhou et al. (2019) noted that dietary supplementation of baicalein improved the growth performance of animals, which associated with the increment of serum GPX concentration. GPX and GSH are the main antioxidant enzymes in mammals that could remove the reactive oxygen species, thus avoiding the occurrence of various kinds of diseases and the alternation of physiology and behavior caused by oxidative stress, thereby benefitting the growth performance (Yuan et al., 2007). In this study, the concentrations of serum GPX ($P = 0.047$) and GSH ($P = 0.074$) in the TRT3 group were higher than TRT1 or CON groups. Furthermore, a positive correlation between supplementation of enzymatic bio-conversion of SB extract in the diet and the concentration of serum GPX and GSH were seen. A similar study was reported by Zeng et al. (2019), pigs consuming antioxidant herbal extract containing diet increased serum GPX concentration thereby improving the growth performance. Su et al. (2018) reported that dietary supplementation of antioxidant herbal extract increased serum GSH concentration of pigs thereby improving the growth performance. Therefore, we speculated that the growth performance improved by dietary supplementation of enzymatic bio-conversion of SB extract was associated with the improvement of antioxidant activities which were shown as the concentrations of serum GSH and GPX increased. In addition, there was no significant difference in antioxidative indicators between TRT1 or CON groups, but the ATTD of DM ($P = 0.095$) in the TRT1 group was higher than CON group. A positive correlation between supplementation of apramycin in the diet and the ATTD of DM was seen. Thus, the growth performance of pigs improved by dietary supplementation of apramycin was considered as the increment of the nutrient digestibility. Despite the mechanism of growth promotion about dietary supplementation of apramycin or enzymatic bio-conversion of SB extract was different, but the results of growth performance promotion were similar. Therefore, the enzymatic bio-conversion of SB extract has the potential to be used as an alternative to antibiotics on the growth performance.

The improvement of the counts of intestinal microbiota was considered as the primary reason for the increment of nutrient

digestibility (Liu et al., 2016; Ju et al., 2019). However, there were no significant differences in fecal microbiota between all diet groups in this study. As reported by An et al. (2010), dietary supplementation of 0.1, 0.3, or 0.5% SB extract did not affect the counts of cecal *Lactobacillus* and *E. coli* in broiler chicks. The SB extract probably without present superb antibacterial effects in the vivo despite its excellent antibacterial effects in the vitro (Lu et al., 2011; Billah et al., 2019). When animals consuming SB extract containing diet, due to the hydrophilic properties of baicalin or wogonoside, they are poorly absorbed in the gastrointestinal tract by its native form and must be hydrolyzed by intestinal microbiota to baicalein or wogonin (Lin et al., 2003; Akao et al., 2009; Kim et al., 2009) thereby enhancing uptake (Lai et al., 2003; Lu et al., 2007). However, the hydrolysis was incomplete and lower than 50% (Trinh et al., 2009). This means more than 50% of flavonoids from SB extract exist in the gastrointestinal and exert biochemistry activity such as antibacterial (Zhao et al., 2018). However, the effects of antibacterial activity by flavonoids of SB extract is dose-dependent (Zhou et al., 2019). In vitro, the dose of SB extract exerts effective antimicrobial activity was average 0.2% (Trinh et al., 2018). We speculated that dietary supplementation of 0.1% SB extract can not produce enormous unabsorbed flavonoids in the gastrointestinal to exert antibacterial activity to change the balance of intestinal microbiota. In addition, due to the baicalin and wogonoside were enzymatic bio-converted by β -glucuronidase to form baicalein and wogonin in the process of enzymatic bio-conversion of SB extract. The baicalein and wogonin were directly absorbed through the small intestine and could not reach the subsequent gastrointestinal to exert biochemistry activity (Lu et al., 2011; Chen et al., 2016). Thus, the fecal microbiota was not affected by dietary supplementation of SB extract or enzymatic bio-conversion of SB extract, thereby insignificantly affecting the nutrient digestibility.

The noxious gas emission was related to the unabsorbed nutrient ingredients fermented by intestinal microbiota in the gastrointestinal, particularly harmful bacteria (Ferket et al., 2002). High levels of fecal *Lactobacillus* and low levels of *E. coli* counts were associated with fecal gas emission reduction (Liu et al., 2020). As reported by Dang et al. (2020), pigs fed a diet supplementing with antioxidant herbal extract decreased fecal *E. coli* counts thereby reducing the NH_3 emission. Liu et al. (2020) observed a reduction fecal NH_3 and H_2S emission which associated with pigs consuming an antioxidant herbal extract containing diet decreased fecal *E. coli* counts and increased fecal *Lactobacillus* counts. In the present study, the fecal microbiota including *Lactobacillus* and *E. coli* were not affected by dietary supplementation of apramycin, SB extract, or enzymatic bio-conversion of SB extract. This was considered as the reason for fecal gas emission insignificantly affected by all diet groups.

Psychologic states produced by environmental or physiologic stresses behave as excessive excitation of the central nervous system (Figueiredo et al., 2007), which associated with the hyper-secretion of cortisol, epinephrine, and norepinephrine (Martínez-Miró et al., 2016). Animals under stress states lead to anxiety (Gold and Chrousos, 2002), thereby causing deleterious physiological changes and alters normal functioning such as refusing to eat, increasing fighting, or decreasing immune states (Segerstrom and Miller, 2004; Lee et al., 2007). In the present study, the concentration of cortisol, epinephrine, and norepinephrine had no significant differences in the TRT1, TRT2, or TRT3 groups. Park et al. (2016) reported that broiler chicks fed with 0.025 or 0.05% SB extract did not affect the concentration of cortisol, epinephrine, and norepinephrine. However, no more comparisons could be made with other studies because investigations on the use of SB extract in pigs are limited. The supplementation of 0.1% SB extract or enzymatic bio-conversion of SB extract did not impair the physiological status.

The concentrations of BUN, creatinine, protein and insulin in blood were used as an indicator of renal function (Waisundara, 2010; Ahad et al., 2014; Long et al., 2016). In the present study, the concentrations of insulin, BUN, creatinine, and protein in blood were not affected by dietary supplementation of apramycin, SB extract, or enzymatic

bio-conversion of SB extract. Moreover, the protecting effect of flavonoids from SB extract on renal histological damage already demonstrated by Lai et al. (2016) and Dai et al. (2017). The results of this study indicated that dietary supplementation with 0.1% SB extract or enzymatic bio-conversion of SB extract had no adverse effects on the renal function in growing pigs.

The oxidative process is an essential mechanism for the production of ATP in aerobic organisms, this means the formation of considerable counts of relative oxygen species, thereby damaging the DNA (Burney et al., 1999), compromising the cellular function (Hawkins and Davies, 2001), and implicating several diseases such as diabetes (Ozsoy et al., 2008) to negatively affect the growth and production of animals (Fejercáková et al., 2014). Oxidative stress will increase the GSSG level along with corresponding decrease in GSH and GPX (Wen et al., 2013; Wang et al., 2019). Increased antioxidant enzyme activities mean to reduce oxidative stress levels (Waisundara, 2010). In the present study, the concentrations of GPX ($P = 0.047$) and GSH ($P = 0.074$) in TRT3 group were higher than CON or TRT1 groups. Despite the antioxidant activity of SB extract has plenty reported (Chan et al., 2010; Li et al., 2015a), but dietary supplementation of SB extract did not affect the antioxidative indicators in this study. A similar study was reported by Liao et al. (2018), they found that broiler chicks fed a diet supplemented with 60, 120, 180, or 240 mg/kg SB extract did not affect the GPX concentration. While Zhou et al. (2019) observed an increase of GPX in broiler chicks consuming baicalein containing diet. Rodriguez et al. (2017) noted that the antioxidant activity of SB extract was depending on the content of flavonoid from SB extract. The antioxidative activities of baicalein (Kang et al., 2012), baicalin (Wang et al., 2013), and wogonin (Lee et al., 2007) are well known. Moreover, baicalin was seen to be increasing the expression of antioxidant enzyme (Waisundara, 2010). Wen et al. (2013) noted that therapy with baicalin will inhibit the formation of GSSG along with increase in GSH level under cadmium chloride administration. Park et al. (2008) reported that baicalin attenuated the hepatic glutathione depletion 24 h after injection of CCl_4 . Thus, increased concentration of GSH or GPX observed in TRT3 group, but not TRT2 group, probably due to the induction enzyme (β -glucuronidase) present in the process of enzymatic bio-conversion of SB extract could bio-transform of low bioavailability flavonoid (baicalein or wogonin) into high bioavailability flavonoid (baicalin or wogonoside), thereby ameliorated antioxidant ability (Lai et al., 2003; Zhang et al., 2005; Kim et al., 2009; Xu and Ji, 2013).

5. Conclusion

The results of this study partially verified our hypothesis, positive effects on growth performance and antioxidative indicators were observed in dietary supplementation of enzymatic bio-conversion of SB extract compared with control group. Moreover, the enzymatic bio-conversion of SB extract has comparable effects with apramycin on growth performance, nutrient digestibility, fecal microbiota, fecal gas emission, and blood hematology, however, the enzymatic bio-conversion of SB extract was better than apramycin in the improvement of antioxidative indicators. Besides, there were no significant differences in all test indexes with dietary supplementation of SB extract. Thus, the enzymatic bio-conversion of SB extract, but not SB extract, at the dose of 0.1% has the potential to be used as an alternative to antibiotics in the diet of animals.

Author declaration

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the

order of authors listed in the manuscript has been approved by all of us.

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

We further confirm that any aspect of the work covered in this manuscript that has involved either experimental animals or human patients has been conducted with the ethical approval of all relevant bodies and that such approvals are acknowledged within the manuscript.

We understand that the Corresponding Author is the sole contact for the Editorial process (including Editorial Manager and direct communications with the office). He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs. We confirm that we have provided a current, correct email address which is accessible by the Corresponding Author and which has been configured to accept email from: inhokim@dankook.ac.kr

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De Xin Dang: Writing - original draft, Investigation, Writing - review & editing. **Yan Jiao Li:** Formal analysis, Investigation. **In Ho Kim:** Conceptualization, Methodology, Supervision, Writing - review & editing.

Declaration of Competing Interest

No potential conflict of interest was reported by the authors.

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